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In [2]: # Program 9: Paired t-test (marks before and after training)
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\diamonds.csv" # <-- change to your file path
BEFORE = None # column name; None = 1st numeric column
AFTER = None # column name; None = 2nd numeric column

df = pd.read_csv(FILE)
num = df.select_dtypes(include="number").columns
if BEFORE is None:
    BEFORE = num[0]
if AFTER is None:
    AFTER = num[1]

data = df[[BEFORE, AFTER]].dropna()
t, p = stats.ttest_rel(data[BEFORE], data[AFTER])

print("Before:", BEFORE, "mean =", data[BEFORE].mean())
print("After :", AFTER, "mean =", data[AFTER].mean())
print("t-statistic:", t)
print("p-value   :", p)

if p < 0.05:
    print("Result: The training has a significant effect.")
else:
    print("Result: No significant effect of the training.")

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Before: carat mean = 0.7979397478680014
After : depth mean = 61.749404894327036
t-statistic: -9460.979445817939
p-value   : 0.0
Result: The training has a significant effect.

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In [5]: # Program 9: Paired t-test (marks before and after training)
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\ecommerce_customer_behavior_dataset.csv" # <-- change to your file path

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BEFORE = None          # column name; None = 1st numeric column
AFTER = None           # column name; None = 2nd numeric column

df = pd.read_csv(FILE)
num = df.select_dtypes(include="number").columns
if BEFORE is None:
    BEFORE = num[0]
if AFTER is None:
    AFTER = num[1]

data = df[[BEFORE, AFTER]].dropna()
t, p = stats.ttest_rel(data[BEFORE], data[AFTER])

print("Before:", BEFORE, "mean =", data[BEFORE].mean())
print("After :", AFTER, "mean =", data[AFTER].mean())
print("t-statistic:", t)
print("p-value   :", p)

if p < 0.05:
    print("Result: The training has a significant effect.")
else:
    print("Result: No significant effect of the training.")

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Before: Age mean = 35.0326
After : Unit_Price mean = 455.83411999999999
t-statistic: -41.756865247240995
p-value   : 0.0
Result: The training has a significant effect.

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In []: